

Performance Analysis of Indoor Visible Light Communication System with Angle Diversity Transmitter

Vipul Dixit
Department of ECE
PDPM-IIITDM, Jabalpur
Jabalpur, India
vipuldixit86@gmail.com

Atul Kumar
Department of ECE
PDPM-IIITDM, Jabalpur
Jabalpur, India
atul.kumar@iiitdmj.ac.in

Abstract— In this paper, angle diversity transmitter (ADT) is implemented with indoor visible light communication (VLC) system, namely, VLC-ADT system. The error performance analysis of VLC-ADT system is presented over the line of sight (LOS) VLC channel in a downlink environment. The bit-error-rate (BER) of indoor VLC-ADT system is analyzed for different receiver locations in the room. The proposed VLC-ADT system reduces the disparity between the minimum and maximum received signal-to-noise ratio (SNR) observed at an optimum semi-angle at half illuminance, $\phi_{1/2}$. The BER performance of indoor VLC-ADT system is also evaluated for the different values of elevation angle considering different locations of the receiver. The simulated results are validated with the analytical results.

Index Terms— Bit-error-rate (BER), Visible light communication (VLC), Angle diversity transmitter (ADT), Signal to noise ratio.

I. INTRODUCTION

Recently, the demand for wireless connected devices has grown-up speedily. The reports show that there will be 13.1 billion wireless devices by the year 2023 [1]. Visible light communication (VLC) has emerged as a promising technology to satisfy this massive demand of wireless devices [2, 3]. The VLC demonstrates the dual use of light emitting diode (LED) for lightening and communication. VLC operates in the huge terahertz spectrum of 430 THz to 790 THz and provides a wide license free spectrum [3, 4]. In addition, VLC offers several other advantages such as security to the threats, low power consumption, small size, long life, no eye hazard and ease of availability, and hence can be considered as a suitable candidate for 5G and beyond communication [3]. VLC finds numerous applications in various fields such as underwater communication, mobile camera communication, light fidelity (Li-Fi), indoor positioning and localization, medical instruments and vehicle to vehicle communication [3, 4].

Angle diversity transmitter (ADT) which uses multiple LEDs aligned at different angles of elevation is a well-established technique in the VLC systems. ADT is used to

improve the indoor localization in VLC system [5]. ADT is adopted to realize the space division multiple access (SDMA) which in result outperforms the optical time division multiple access (OTDMA) [6]. ADT based SDMA scheme provides an improvement of 26% in spectral efficiency if 37 LEDs are incorporated in ADT structure [7, 8]. In [9], VLC system with ADT provides improvement in maximum data rate and quality of service (QoS). The signal to noise ratio (SNR) distribution of VLC system with ADT is analyzed for wide field of view (W-FOV) receiver and angle diversity receiver (ADR) [10].

The reported works in literature focuses on the improvement in achievable data rate and uniformity of illuminance in the room [5-10]. However, for the practical implementation of any communication system, error-free transmission of data is a major concern. Thus, in this paper, the bit-error-rate (BER) performance of a VLC system with ADT is analyzed.

In this paper, ADT is implemented with indoor VLC system to improve the BER performance. The proposed indoor VLC-ADT reduces the disparity between minimum and maximum received SNR observed at an optimum value of semi angle at half illuminance, $\phi_{1/2}$. The BER performance of the indoor VLC-ADT system is analysed for the different receiver locations in the room and at different elevation angles. The simulation results are validated with the analytical results.

The paper is outlined as follows: In Section II, the system model of indoor VLC-ADT system is presented. The performance analysis of the proposed VLC-ADT system is presented in Section III and the results are discussed in Section IV. The paper is finally concluded in Section V.

II. SYSTEM MODEL

A downlink indoor VLC system based on intensity modulation and direct detection (IM/DD) with multiple LED's as transmitter and a photo detector (PD) as receiver is under consideration. A room model with the dimension of (x, y, h) is considered with ADT located at the center position of the roof, i.e., $T(x_i, y_i, z_i)$, and the receiver is located with the co-

ordinates $R(x_r, y_r, z_r)$, where z_r is the height of receiver from the ground.

The user is assumed to be static in nature. The power of received signal through non line of sight (NLOS) channel in VLC systems is much lower than that of line of sight (LOS) channel. Therefore, this paper considers only LOS channel for the purpose of communication between the transmitter and receiver [11-12].

A. Angle Diversity Transmitter (ADT)

An ADT setup consists of N_t LEDs is under consideration when a LED is placed at the center and remaining $(N_t - 1)$ side LEDs are placed at a certain angle of inclination with respect to the central LED as shown in Fig. 1. All the co-ordinates and orientations of the LEDs in ADT structure are carefully selected to avoid the dark zones of lighting and communication. The azimuth and elevation angles of the PD are assumed to be 0° and 90° , respectively. The central LED in the ADT structure is located at the co-ordinates of (x_t, y_t, z_t) with azimuth and elevation angle of 0° and 90° , respectively. The azimuth and elevation angles of all $(N_t - 1)$ side LED's are given as

- The azimuth angle of the $(N_t - 1)$ side LED's are given by

$$\alpha_i = \begin{cases} 0^\circ, & \text{for } i = 1 \\ \frac{2\pi(i-1)}{N_t - 1}, & \text{for } i = 2, \dots, N_t - 1 \end{cases}$$

- The elevation angle of $(N_t - 1)$ side LEDs are equal, i.e., $\beta_i = \beta$, for $i = 1, \dots, N_t - 1$.

For the calculation of elevation angle ϕ , we have considered a point L with co-ordinates (x_l, y_l, z_l) located 1 m below to the normal axis of the transmitting LED as shown in Fig.1. The co-ordinates of point L can be calculated using transmitter location co-ordinates $T(x_t, y_t, z_t)$, azimuth and elevation angles. The co-ordinates of point L can be given as [10]

$$x_l = x_t + \frac{\cos \alpha_i}{\tan \beta_i} \quad (1)$$

$$y_l = y_t + \frac{\sin \alpha_i}{\tan \beta_i} \quad (2)$$

$$\text{and} \quad z_l = z_t - 1 \quad (3)$$

Using these co-ordinates of point L , the angle of emission ϕ can be calculated as [10]

$$\cos \phi = \frac{|LT|^2 + |RT|^2 - |LR|^2}{2|LT||RT|}, \quad (4)$$

where

$$|RT| = \sqrt{(x_r - x_t)^2 + (y_r - y_t)^2 + (z_r - z_t)^2} \quad (5)$$

$$|LT| = \sqrt{1 + d^2} \quad (6)$$

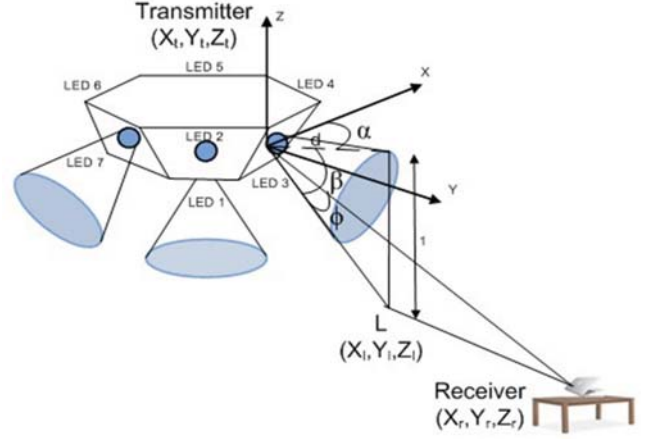


Fig.1. System model of indoor VLC- ADT system.

$$\text{and} \quad |LR| = \sqrt{(x_r - x_l)^2 + (y_r - y_l)^2 + (z_r - z_l)^2} \quad (7)$$

The distance, d , between the transmitter T and point L can be given as

$$d = \frac{1}{\tan \beta_i} \quad (8)$$

III. PERFORMANCE ANALYSIS

The received signal at the user is given as

$$y = \sum_{i=1}^{N_t} R_p h_i 2P_i^{avg} x + n, \quad (9)$$

where R_p is the responsivity of the PD, h_i is the LOS channel gain of i^{th} transmitting LED, x is on-off keying (OOK) modulated signal, n is the noise present at the receiving PD, $P_i^{avg} = P_t / N_t$ is the transmitted optical power per LED, and P_t is the total transmitted power from the ADT structure. To retain the average transmitted optical power as P_i^{avg} , the factor 2 is included in (9) as there is no optical power transmission for bit '0'.

The LOS channel gain between the i^{th} transmitting LED and receiver is given by [4]

$$h_i = \begin{cases} \frac{(m+1)AR_p}{2\pi d_i^2} \cos^m(\phi_i) T(\psi_i) g(\psi_i) \cos(\psi_i), & \text{for } \psi_i \leq \psi_c \\ 0, & \text{otherwise} \end{cases} \quad (10)$$

where R_p is the responsivity of the photodetector, A is the active area of the receiving PD, d_i is the distance between the i^{th} transmitting LED and receiver, ϕ_i is the angle of emission from the i^{th} transmitting LED, ψ_i is angle of incidence at the receiver from the i^{th} transmitting LED, $g(\psi_i) = n^2 / \sin^2 \psi_c$ is the non-imaging optical concentrator gain with refractive index n , $T(\psi_i)$ is the optical filter gain, ψ_c is field of view (FOV), $m = -\ln 2 / \ln(\cos \phi_{1/2})$ is Lambertian order of transmitting LED and $\phi_{1/2}$ is the semi-half angle of illuminance.

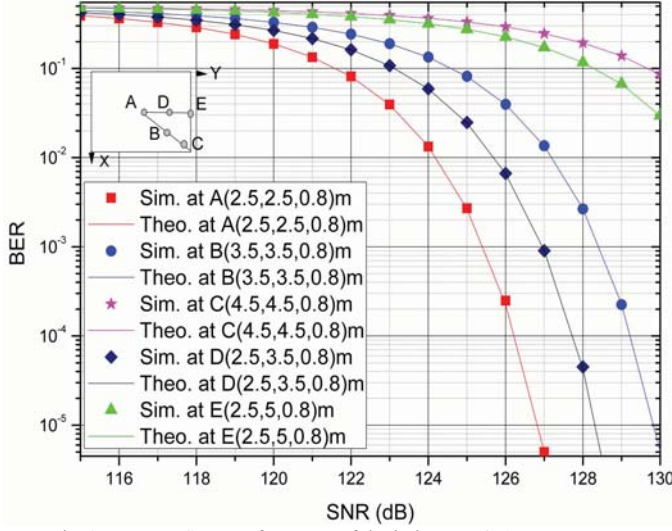


Fig. 3. BER vs. SNR performance of the indoor VLC-ADT system.

The noise n is the additive white Gaussian noise (AWGN) at the receiver with zero mean and variance of σ_n^2 . The n comprises of shot and thermal noise, the total variance of n can be given as [4, 13]

$$\sigma_n^2 = \sigma_{shot}^2 + \sigma_{thermal}^2, \quad (11)$$

where σ_{shot}^2 and $\sigma_{thermal}^2$ is the variance of shot and thermal noise, respectively. The variance of shot noise is defined as [4, 13]

$$\sigma_{shot}^2 = 2qI_{bg}I_2B + 2qRP_iBh_i, \quad (12)$$

where I_{bg} is the background current whose typical values is $5100 \mu A$ under direct exposure to sunlight, q is the electronic charge, B is the equivalent noise bandwidth which is equal to the modulation bandwidth, and $I_2 = 0.562$ is the noise bandwidth factor. Similarly, the variance of thermal noise is defined as [4, 13]

$$\sigma_{thermal}^2 = \frac{8\pi kT_k A I_2 B^2 \eta}{G} + \frac{16\pi^2 kT_k A^2 I_3 B^3 \eta^2 \Gamma}{g_m}, \quad (13)$$

where k is the Boltzmann's constant, Γ is the FET channel noise factor, G is the open loop voltage gain, T_k is the absolute temperature, g_m is the FET transconductance, η is the fixed capacitance of PD per unit area, and $I_3 = 0.0868$. The following values are used in this paper to model the thermal noise: $G = 10$, $T_k = 295K$, $\Gamma = 1.5$, $g_m = 30mS$, and $\eta = 112pF/cm^2$.

The SNR for VLC system using the ADT is given by

$$SNR_{VLC-ADT} = \frac{\text{Total received signal power } (P_r)}{\text{Noise power } (\sigma_n^2)} \quad (14)$$

The total received signal power, P_r , is expressed as

$$P_r = \sum_{i=1}^{N_i} h_i P_i^{avg} \quad (15)$$

Thus, the BER of OOK modulated VLC system using ADT is given by

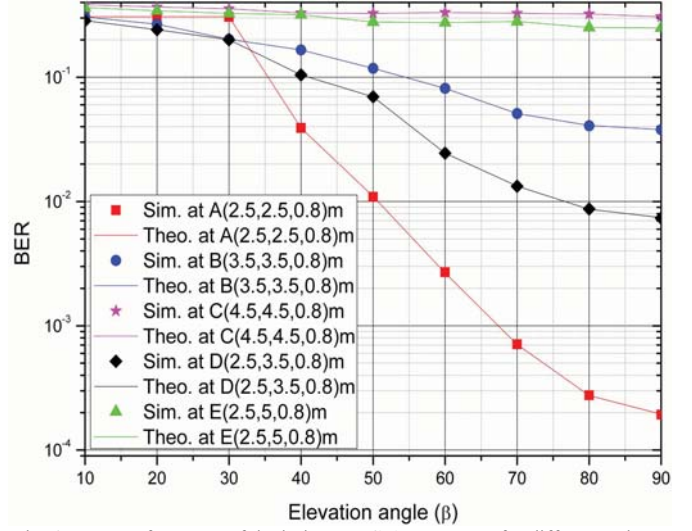


Fig. 4. BER performance of the indoor VLC-ADT system for different values of elevation angle.

$$BER = Q\left(\sqrt{SNR_{VLC-ADT}}\right) \quad (16)$$

Substituting (14), (11) and (15) in (16), the BER of OOK modulated VLC system using ADT can be evaluated.

IV. RESULT AND DISCUSSION

In this paper, seven LEDs are considered in the ADT structure, i.e., $N_i = 7$, to simulate the system model. The azimuth angles of the LEDs are assumed as $0^\circ, 0^\circ, 60^\circ, 120^\circ, 180^\circ, 240^\circ$ and 360° , and the elevation angle of side LEDs is fixed at 60° , unless otherwise stated. All the LEDs in the ADT setup is assumed to follow a Lambertian order of $m = 1$ at $\phi_{1/2} = 60^\circ$. The assumed room co-ordinates and simulation parameters for VLC-ADT system are given in Table I. It is to be noted that the channel gain of LOS-VLC system is in the order of 10^{-5} and hence, the SNR exhibits an offset of about 100 dB.

TABLE I
SIMULATION PARAMETERS FOR VLC-ADT SYSTEM

Parameter	Value
Room dimension	$(5 \times 5 \times 3) m^3$
Source	$(2.5, 2.5, 3) m$
Receiver	$A(2.5, 2.5, 0.8) m$ $B(3.5, 3.5, 0.8) m$ $C(4.5, 4.5, 0.8) m$ $D(2.5, 3.5, 0.8) m$ $E(2.5, 5, 0.8) m$
Azimuth angle (α_i)	$0^\circ, 0^\circ, 60^\circ, 120^\circ, 180^\circ, 240^\circ, 360^\circ$
Elevation angle (β_i)	$90^\circ, 60^\circ, 60^\circ, 60^\circ, 60^\circ, 60^\circ, 60^\circ$
FOV	60°
Responsivity (R_p)	1
Active area of PD (A_r)	$1 cm^2$
Refractive Index (n)	1.5
Optical filter Gain (T_s)	1
Noise Bandwidth (B)	50 MHz
LED's Semi-half Angle ($\Phi_{1/2}$)	60°
Transmission rate (R_b)	50 Mbps

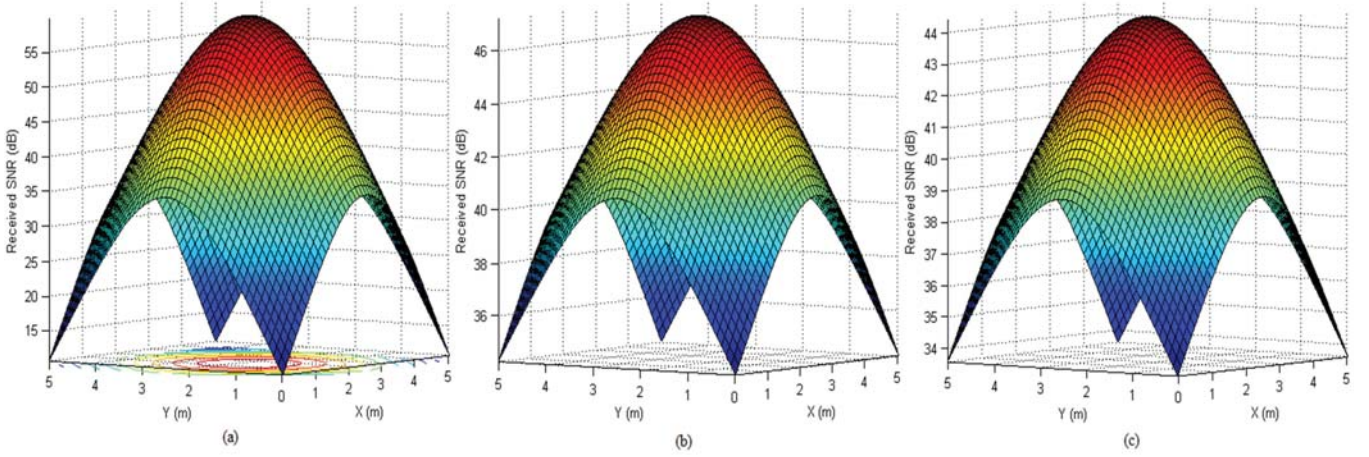


Fig. 5. Received SNR distribution curves of indoor VLC-ADT system for different values of semi-half angle (a) $\phi_{1/2} = 20^\circ$ (b) $\phi_{1/2} = 60^\circ$ and (c) $\phi_{1/2} = 80^\circ$.

In Fig. 3, the BER vs. SNR curves of indoor VLC-ADT system are plotted for different locations of receiving PD. In this paper, five different locations of the receiving PD, i.e., A, B, C, D and E, are considered with the co-ordinates (2.5, 2.5, 0.8) m, (3.5, 3.5, 0.8) m, (4.5, 4.5, 0.8) m, (2.5, 3.5, 0.8) m and (2.5, 5, 0.8) m, respectively. The figure shows that the BER performance of the indoor VLC-ADT system improves as SNR increases. Further, the figure demonstrates that the BER performance degrades as receiver moves away from A, i.e., center location, to E, i.e., corner location of the room. The reason behind this is that as the user moves away from the central location, the LOS channel conditions become worse.

The BER performance of indoor VLC-ADT system is plotted for the different values of elevation angle β at five locations of the receiving PD in Fig. 4. The SNR is kept fixed at 125 dB for this analysis. The figure shows that the BER performance improves for the elevation angle greater than 30° at all of the five different locations of the receiving PD. Further, the figure shows that the improvement in BER performance is greater for the central locations of PD in comparison to the corner locations. The reason behind this is that more number of LOS channel path will be present between the ADT and the receiving PD, when the elevation angle is increased above 30° at certain receiver locations. The simulation results are perfectly matching with the analytical results.

The Fig. 5 compares the SNR distribution curves of the indoor VLC-ADT system for different values of semi-half angle of illuminance, $\phi_{1/2}$. For the comparison of SNR distribution curves, three different values of $\phi_{1/2}$ are considered, i.e., 20° , 60° and 80° . The figure shows that the SNR distribution curve of the indoor VLC-ADT system becomes more uniform as the value of semi-half angle of illuminance, $\phi_{1/2}$ increases. The difference between the minimum and maximum value of received SNR is least for the $\phi_{1/2} = 80^\circ$ as shown in Fig. 5 (c). Table II gives a detailed comparison between the minimum, maximum and average values of the received SNR of the indoor VLC-ADT system for $\phi_{1/2} = 20^\circ$, 60° and 80° .

TABLE II
RECEIVED SNR VALUES FOR INDOOR VLC-ADT SYSTEM

Value of $\phi_{1/2}$	SNR Values		
	Min. Value	Max. Value	Avg. Value
20°	10.7173	59.8472	35.28225
60°	34.2738	47.1706	40.7222
80°	33.6036	44.4253	39.01445

V. CONCLUSION

In this paper, an ADT setup is implemented with the indoor VLC system for the performance analysis. The BER performance of the indoor VLC-ADT system consisting of seven LEDs is analyzed for the different locations of the receiving PD. The simulated results are validated with the analytical results. It is concluded that the BER of indoor VLC-ADT system degrades as receiver moves away from center location to corner location of the room. Further, the BER performance improves for the elevation angle greater than 30° for all of the five different locations of the receiving PD. Furthermore, the SNR distribution curve of the indoor VLC-ADT system becomes more uniform as the value of semi-angle at half illuminance, $\phi_{1/2}$, increases.

REFERENCES

- [1] "Cisco Annual Internet Report, 2018-2023 white paper", Cisco, San Jose, CA, USA, 2020.
- [2] D. C. O'Brien et al., "Visible light communications: Challenges and possibilities," in *Proc. IEEE PIMRC*, Cannes, France, 2008, pp. 1–5.
- [3] H. Le Minh, Z. Ghassemlooy, D. O'Brien and G. Faulkner, "Indoor Gigabit Optical Wireless Communications: Challenges and Possibilities", in *Proc. 12th ICTON*, Munich, 2010, pp. 1-6.
- [4] Z. Ghassemlooy, W. Popoola, and S. Rajbhandari, *Optical Wireless Communications: System and Channel Modeling with MATLAB*, 1st ed., New York: CRC press, 2013.
- [5] L. Yin, X. Wu and H. Haas, "Indoor Visible Light Positioning with Angle Diversity Transmitter", 2015 IEEE 82nd Vehicular Technology Conference (VTC2015-Fall), Boston, MA, 2015, pp. 1-5.
- [6] Z. Chen and H. Haas, "Space division multiple access in visible light communications", *International Conference on Communications (ICC)*, IEEE, 2015, pp. 5115-5119.

- [7] Z. Chen, D. A. Basnayaka and H. Haas, "Space division multiple access in optical attocell networks", 2016 IEEE Wireless Communications and Networking Conference, Doha, 2016, pp. 1-5.
- [8] Z. Chen, D. A. Basnayaka and H. Haas, "Space Division Multiple Access for Optical Attocell Network Using Angle Diversity Transmitters", in *Journal of Lightwave Technology*, vol. 35, no. 11, pp. 2118-2131, 1 June1, 2017.
- [9] Y. S. Eroglu, I. Güvenç, A. Sahin, Y. Yapici, N. Pala and M. Yüksel, "Multi-Element VLC Networks: LED Assignment, Power Control, and Optimum Combining", in *IEEE Journal on Selected Areas in Communications*, vol. 36, no. 1, pp. 121-135, Jan. 2018.
- [10] S. H. Younus, A. A. Al-Hameed, A. T. Hussein, M. T. Alresheedi and J. M. H. Elmirghani, "Multi-Branch Transmitter for Indoor Visible Light Communication Systems", 2020 22nd International Conference on Transparent Optical Networks (ICTON), Bari, Italy, 2020, pp. 1-6.
- [11] L. Zeng, D. O'Brien, H. Minh, G. Faulkner, K. Lee, D. Jung, Y. Oh, and E. T. Won, "High Data Rate Multiple Input Multiple Output (MIMO) Optical Wireless Communications Using White LED Lighting," *IEEE J. Sel. Areas Commun.*, vol. 27, no. 9, Dec. (2009), 1654–1662.
- [12] V. Dixit and A. Kumar, "Performance analysis of non-line of sight visible light communication systems", *Optics Communication*, vol. 459, 125008, 2020.
- [13] T. Komine and M. Nakagawa, "Fundamental analysis for visible-light communication system using LED lights," *IEEE Trans. Consum. Electron.*, vol. 50, no. 1, pp. 100-107, Feb. 2004.